higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N6

20 NOVEMBER 2019

This marking guideline consists of 10 pages.

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QUESTION 1

1.1 Calculate the number of bolts for the part marked T:

Force = area eff × stress × number

Stress = Pv = 100 MPa

$$N = \frac{Force}{Pv \times Aeff}$$ Cl 9.5.1

$$Aeff = \frac{\pi(Dia - 0{,}9382 \times P)^2}{4}$$

$$Aeff = \frac{\pi(16 - 0{,}9382 \times 2)^2}{4}$$ ✓

Aeff = 156,67 mm² ✓

$$N = \frac{68 \times 10^3}{100 \times 156{,}67}$$ ✓✓

N = 4,34 bolts ✓

Use 5M16 bolts. ✓ (6)

1.2 Select a suitable equal-leg angle for the part marked S:

The total length of part P = 1,48 metres (1 480 mm)

Force = 60 kN (strut)

Select a trial section:

60 × 60 × 8 rolled steel angle

Area = 0,903 × 10^3 mm² ✓

R_{min} = 11,6 mm ✓

L_{eff} = 1 480 × 0,85 ✓

= 1 258 mm ✓ SABS 0162 Cl 8.2.1

Calculate slenderness ratio:

$$L/r = \frac{1\ 258}{11{,}6} = 108{,}5$$ ✓ Table 17 SABS 0162

108,5 = 70,5 MPa ✓

Then: Force = stress × area

Force = 70,5 N/mm² × 0,903 × 10^3 ✓

Force = 63, 66 kN (suitable) ✓ (8)

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QUESTION 2

Calculate the direct shear on each bolt:

Force direct = force/number of bolts

$F_{direct} = 18\ kN/4 = 4,5\ kN$✓

Calculate the distance from the centroid to the furthest bolt:

$r = \sqrt{x^2+y^2}$

$r = \sqrt{137,5^2+62,5^2}$✓✓

$r = 151,04\ mm$✓

The direct load on the bolts due to the imposed load:

$\Sigma cwm = \Sigma acwm$

$(18 \times 320) = (F_T \times 151,04 \times 4)$✓✓

$$F_T = \frac{18 \times 320}{151,04 \times 4}$$

$F_T = 9,53\ kN$✓

The resultant load on each bolt:

$$F_R = \sqrt{(Fs^2 + Ft^2) + (2\ Ft\ Fs\ \cos\ \phi)}$$✓

$$F_R = \sqrt{(4,5^2 + 9,53^2)+\left(2 \times 9,53 \times 4,5 \times \cos\frac{137,5}{151,04}\right)}$$

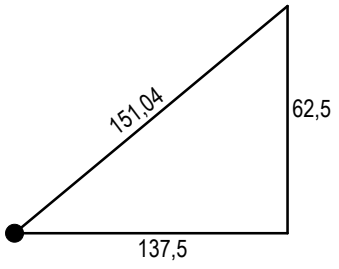
$F_R = \sqrt{(20,25+90,82)+78,16}$ ✓✓

$F_R = 13,76\ kN$✓

Alternate method for the resultant load on each bolt:

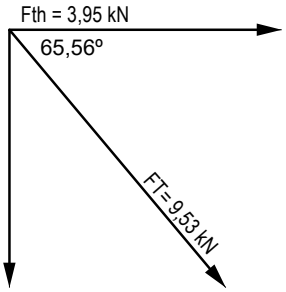
The resultant load on each bolt:

$$Tan\phi = \frac{137,5}{62,5} = 65,56°$$✓



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<u>The resultant load on each bolt:</u>



$F_{Thor} = Ft \times \cos 65,56°$

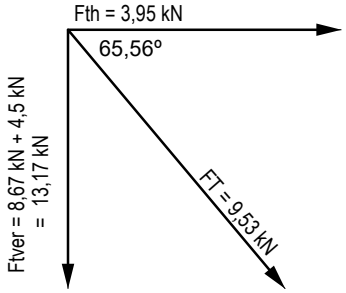
$= 9,53 \times 0,414$

$F_{Thor} = 3,95$ kN✓

$F_{Tvert} = Ft \times \sin 65,56°$

$= 9,53 \times 0,91$

$F_{Tvert} = 8,67$ kN + 4,5 kN = 13,17 kN✓



<u>Resultant force:</u>

$F_R = \sqrt{3,95^2 + 13,17^2}$

$F_R = 13,75$ kN✓

<u>Calculate the size of the bolts:</u>

F_R = shear stress × area of bolt

$$\frac{\pi \ d^2}{4} = \frac{Fr}{Shear \ \ stress}$$✓

$$d = \sqrt{\frac{Fr \ \times \ 4}{\pi \ \times \ stress}}$$

$$d = \sqrt{\frac{13,75 \ \times \ 10^3 \ \times \ 4}{\pi \ \times \ 90}}$$✓

d = 13,95 mm✓

Use 4M16 bolts.✓ **[15]**

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QUESTION 3

All references are taken from SANS 10100-1 (2000).

3.1 Calculate the total design dead load of the slab:

Design dead load = volume × density × 9,81 × 10^{-3} × 1,2 Gn Cl 4.2.2.1
= 8 × 4 × 0,3 × 2 400 × 9,81 × 10^{-3} × 1,2 Gn✓
= 271,23 kN✓

Design imposed load = 8 kN/m^2 × 8 m × 4 m × 1,6 Qn✓
= 406,6 kN✓ Cl 4.2.2.1

Total design load = 271,23 + 406,6 = 677,83 kN✓ (5)

3.2 The load to be supported at A (half of the slab):

$\frac{677,83\ kN}{2}$✓

= 338,92 kN✓ (load to be carried by one I-beam)

Calculate bending moment maximum:

$BM_{max} = \frac{WL}{8}$

$BM_{max} = \frac{338,92 \times 4}{8}$ ✓

BM_{max} = 169,46 kNm✓

Calculate section modulus:
Bending stress = 162 MPa

$Z = \frac{BM}{Bending\ stress}$

$Z = \frac{169,46 \times 10^6}{162}$ ✓

Z = 1 046,05 × 10^{-6} m^3✓

Select 610 × 229 × 140 kg/m.✓ (Ze = 1 117 × 10^{-6} m^3) (7)
[12]

QUESTION 4

4.1 Calculate bending moment:

$$BM = \frac{WL}{4} + \frac{Wl^2}{8}$$

$$BM = \frac{102 \times 7}{4} + \frac{40 \times 7^2}{8} \checkmark\checkmark$$

$$BM = 178{,}5 + 245$$

$$BM = 423{,}5 \text{ kNm} \checkmark$$

Calculate section modulus (Z):

$$Z = \frac{M}{f}$$ Cl 4.3.3.4

$$Z = \frac{423{,}5 \times 10^6}{163} \checkmark$$

$$Z = 2\,598\,159{,}5 \text{ mm}^3$$

$$Z = 2\,598{,}16 \times 10^{-6} \text{ m}^3 \checkmark$$

Select 533 × 210 × 122 kg/m.✓ ($Z_e = 2\,798 \times 10^{-6}$ m^3)✓ (7)

4.2 Include the self-weight of the selected steel beam:

Self-weight = mass × 9,81 × 10^{-3}

= 122 kg/m × 9,81 × 10^{-3}✓

s/w = 1,2 kN/m✓

Total UDL = 40 kN/m + 1,2 kN/m = 41,2 kN/m✓

Calculate the revised bending moment max:

$$BM = \frac{WL}{4} + \frac{Wl^2}{8}$$

$$BM = \frac{102 \times 7}{4} + \frac{41{,}2 \times 7^2}{8} \checkmark$$

$$BM = 178{,}5 + 252{,}35$$

$$BM = 430{,}85 \text{ kNm} \checkmark$$

Calculate new section modulus (Z):

$$Z = \frac{M}{f}$$

$$Z = \frac{430{,}85 \times 10^6}{163}$$

$$Z = 2\,643\,251{,}53 \text{ mm}^3$$

$Z = 2\,643,25 \times 10^{-6}\ m^3$✓

The value of Z is less than the Ze-value of the chosen beam, 533 × 210 × 122 kg/m. The beam can therefore be deemed as adequate.✓ (8)

[15]

QUESTION 5

All references are taken from SANS 10100-1 (2000).

Fcu = 25 MPa	Table 2
Fy = 250 MPa	Cl 4.1.5.1
Span = 7,00 metres	Table 3
Density of reinforced concrete: 2 400 kg/m^3	Cl 4.1.5.2
	Cl 4.3.1.2

5.1 Determine the effective depth of the slab:

Effective depth = span/16 — Table 10

Eff d = 5 500/16 — Cl 4.3.6.2.1

Eff d = 343,75 mm✓

Determine the overall depth:

Assume R20 main steel, R8 binders and 25 mm cover. — SABS 0144 Cl 6.1 and SABS 0100-2 Cl 8.2

Overall depth = 343,75 + 10 + 8 + 25✓

Overall depth = 386,75 mm (Use overall depth = 400 mm.)✓

Determine the design dead loads of the beam:

Design dead load = volume × density × 9,81 × 10^{-3} × 1,2 Gn — Cl 4.2.2.1

= 0,40 × 0,27 × 1 × 2 400 × 9,81 × 10^{-3} × 1,2✓

= 3,05 kN/m✓

Design imposed UDL = 3,5 kN/m × 1,6 Qn = 5,6 kN/m✓ — Cl 4.2.2.1

Calculate bending moment maximum:

$$BM_{max} = \frac{WL^2}{8} + \frac{WL^2}{8}$$

$$BM_{max} = \frac{3,05 \times 5,5^2}{8} + \frac{5,6 \times 5,5^2}{8}$$ ✓✓

$$= 11,53 + 21,175$$

BM_{max} = 32,71 kNm✓

Calculate value for K:

$$K = \frac{BM}{Fcu\ b\ d^2}$$ — Cl 4.3.3.4.1

$$K = \frac{32,71 \times 10^6}{25 \times 270 \times 386,75^2} \checkmark$$

$K = 0,03 < K^1 = 0,156 \checkmark$

Provide tension reinforcement only.✓

Calculate distance of lever arm (Z):

$$Z = d\left\{0,5 + \sqrt{0,25 - \frac{k}{0,9}}\right\} \leq 0,95\text{d}$$ Cl 4.3.3.4.1

$$Z = 386,75\left\{0,5 + \sqrt{0,25 - \frac{0,03}{0,9}}\right\} \leq 0,95 \times 386,75\checkmark$$

$$Z = 386,75\{0,965\} \geq 367,4$$

$$Z = 373,4 \ mm > 367,4 \text{ mm}\checkmark$$ (14)

5.2 Calculate tension reinforcement:

$$\text{As} = \frac{M}{0,87 \times Fy \times z}$$

$$\text{As} = \frac{32,71 \times 10^6}{0,87 \times 250 \times 373,4}\checkmark$$ Cl 4.3.3.4.1

$\text{As} = 402,7 \text{ mm}^2$

Use 4R12. ($\text{As} = 452 \text{ mm}^2$)✓

Minimum distance between bars is correct. Cl 11.4.8 (2)

5.3 Check for minimum main reinforcement:

$$\frac{100 \ As}{Ac} = \frac{100 \times 425}{400 \times 270}\checkmark$$ Table 23 Cl 4.11.4

$= 0,39$

$0,39 > 0,8\checkmark$

The minimum reinforcement is not sufficient.

Check for maximum area of reinforcement:

4% of AC Cl 4.11.5.1

$= 4\% \times 400 \times 270\checkmark$

$= 5\ 120 \text{ mm}^2\checkmark$

$425 < 4\ 320$

The maximum reinforcement is sufficient. (4)

[20]

QUESTION 6

All references are taken from SANS 10100-1 (2000).

Fcu = 30 MPa Table 2 Cl 4.1.5.1
Fy = 450 MPa Table 3 Cl 4.1.5.2
Axial load = 3 220 kN

6.1 Calculate the number of longitudinal reinforcement:

$N = 0{,}4\ Fcu\ Ac + 0{,}67\ Fy\ Asc$ Cl 4.7.4.3

$3\,220 \times 10^3 = 0{,}4 \times 30 \times \left(\frac{\pi\ 525^2}{4}\right) + 0{,}67 \times 450 \times Asc$✓✓

$3\,220 \times 10^3 = 2\,597\,704{,}43 + 301{,}5\ Asc$✓

$3\,220 \times 10^3 - 2\,597\,704{,}43 = 301{,}5\ Asc$

$Asc = \frac{622\,295{,}57}{301{,}5}$✓

$Asc = 2\,063{,}998\ mm^2$✓

Use 8Y20. ($As = 2\,514\ mm^2$) **OR** Use 6R25. ($As = 2\,945\ mm^2$)✓✓ (7)

6.2 Maximum and minimum percentage of the steel reinforcement:

Maximum: 6% of gross sectional area Cl 4.11.4.5.2

$$6\% \times \left(\frac{\pi\ 525^2}{4}\right) = 12\,988{,}52\ mm^2$$✓✓

Maximum: 0,4% of gross sectional area Cl 4.11.4.5

$$0{,}4\% \times \left(\frac{\pi\ 525^2}{4}\right) = 856{,}9\ mm^2$$✓✓ (4)

6.3 Calculate the diameter of the helical binder:

Binders: ¼ of the smallest main bar Cl 4.11.4.5

¼ × 20 = 5 mm (not available)✓

Use R8 binders.✓ (R6 not available)

Calculate the pitch of the binder:

Spacing of binders:

12 × diameter of smallest main bar Cl 4.11.4.5

12 × 20 = 240 mm (maximum)✓✓

Use spacings of 200 mm. (4)

[15]

QUESTION 7

All references are taken from SANS 10100-1 (2000).

Fcu = 25 MPa Table 2 Cl 4.1.5.1
Fy = 450 MPa Table 3 Cl 4.1.5.2

Calculate value for K:

$$K = \frac{BM}{Fcu\ b\ d^2}$$ Cl 4.3.3.4.1

$$K = \frac{285 \times 10^6}{25 \times 1155 \times 525^2}$$

$K = 0,036 < k^1 = 0,156$✓

Calculate the lever arm distances:

$$Z = d\left\{0,5 + \sqrt{0,25 - \frac{k}{0,9}}\right\} \leq 0,95d$$ Cl 4.3.3.4.1

$$Z = 525\left\{0,5 + \sqrt{0,25 - \frac{0,039}{0,9}}\right\} \leq 0,95 \times 525$$✓

$Z = 525(0,955) \leq 0,95 \times 525$✓

$Z = 501,17\ mm \geq 498,75$

Use $Z = 498,75$ mm✓

Calculate area of reinforcement:

$$As = \frac{BM}{Fy\ 0,87\ Z}$$ Cl 4.3.3.4.1

$$As = \frac{285 \times 10^6}{450 \times 0,87 \times 498,75}$$

$As = 1\ 459,59\ mm^2$✓

Use 3Y25 bars. ($As = 1\ 473\ mm^2$)✓ **[9]**

TOTAL: 100

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